

Neural quantum states for the simulation quantum many-body systems

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CECAM school: Machine Learning-Assisted Sampling

Zaragoza, June 18, 2025

$$\psi(\mathbf{X}) = f_{\theta} \left(\begin{array}{c} \text{Diagram of a graph with 5 nodes and green edges. Two nodes at the top are labeled } x_i \text{ and } x_j. A red starburst shape is centered between them, labeled } V_{ij}. \end{array} \right)$$

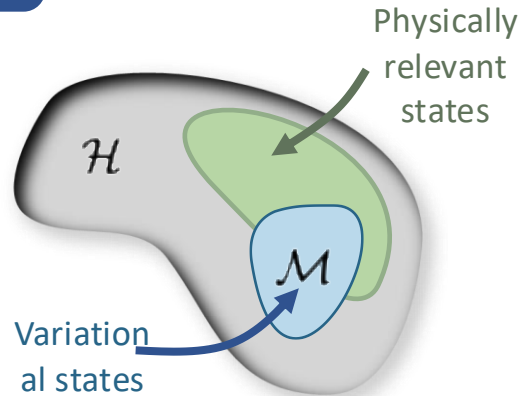
Recap

Variational approach

$$|\psi\rangle = \sum_x \psi(x) |x\rangle$$

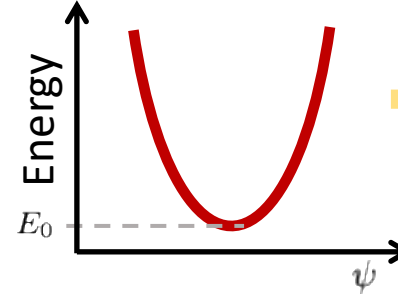
$$|\psi_\theta\rangle = \sum_x \psi_\theta(x) |x\rangle$$

$$\mathcal{M} := \{|\psi_\theta\rangle \mid \theta \in \mathbb{R}^p\}$$



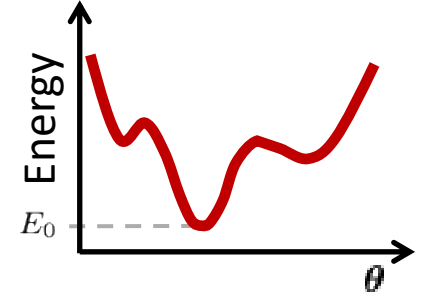
Variational principle

$$|\psi\rangle_{\text{GS}} = \arg \min_{\psi \in \mathcal{H}} E[\psi]$$



$$\psi_{n+1} = \psi_n - \eta \frac{\delta}{\delta \psi} E[\psi]$$

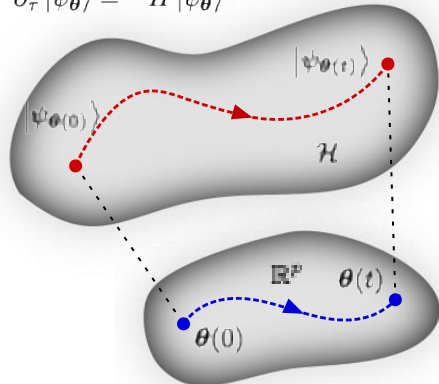
$$\theta_{\text{GS}} = \arg \min_{\theta \in \mathbb{R}^p} E_\theta$$



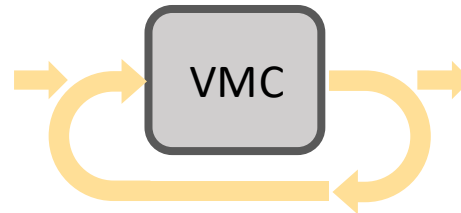
$$\theta_{n+1} = \theta_n - \eta \partial_\theta E_\theta$$

"Stochastic reconfiguration"

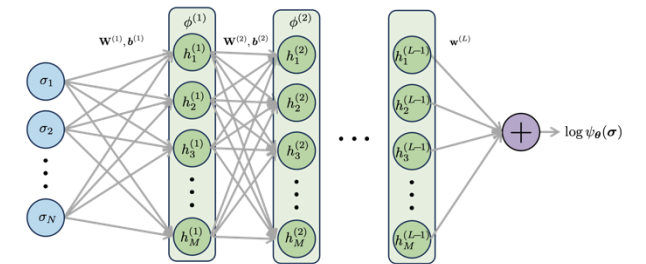
$$\partial_\tau |\psi_\theta\rangle = -\hat{H} |\psi_\theta\rangle$$



Variational Monte Carlo



Neural quantum states

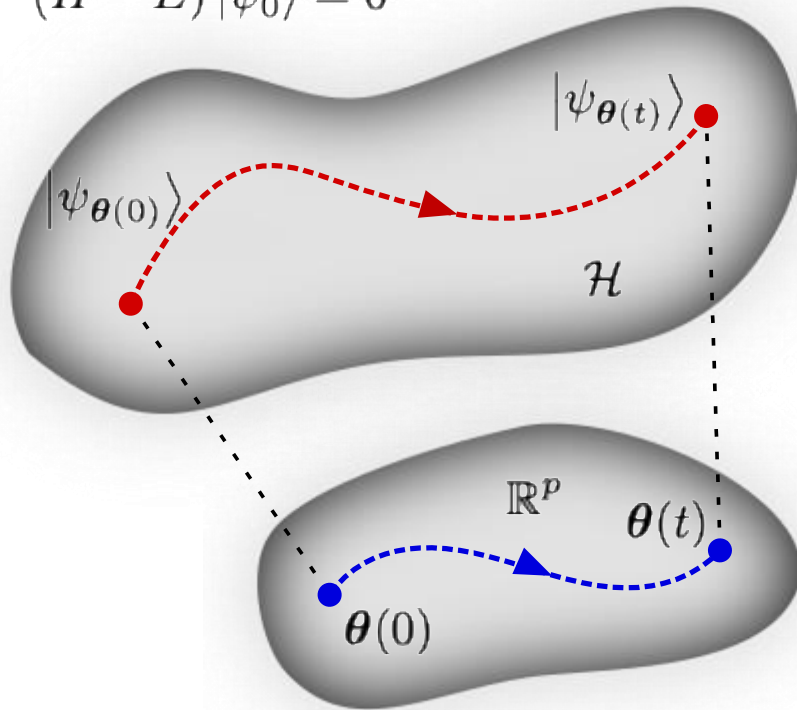


What makes a good NQS?

Sidenote

$$\partial_\tau |\psi_\theta\rangle = -\hat{H} |\psi_\theta\rangle$$

$$-(\hat{H} - E) |\psi_0\rangle = 0$$



?

Variational principle

$$\dot{\theta} = \arg \min_{\dot{\eta}} d_{\text{FS}} \left(e^{-\hat{H}\delta\tau} |\psi_\theta\rangle, |\psi_{\theta+\delta\tau\dot{\eta}}\rangle \right)$$



Stochastic-reconfiguration (SR) equation

$$\dot{\theta} = \mathbf{S}^{-1} \mathbf{F}$$

$$S_{kk'} = \overline{(O_k^* - \overline{O_k^*})(O_{k'} - \overline{O_{k'}})}$$

Quantum geometric tensor

$$F_k = \overline{(O_k^* - \overline{O_k^*})(E_{\text{loc}} - \overline{E_{\text{loc}}})}$$

Forces $\sim$ energy gradient

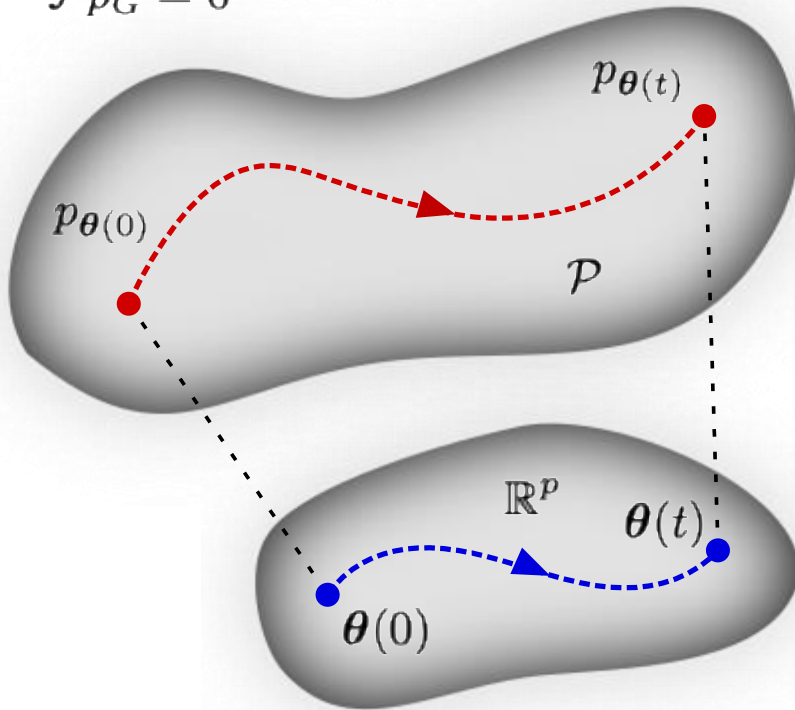
SR rescales the gradient according to the physical effect of each parameter on the wavefunction.

$$\theta_{n+1} = \theta_n - \eta \partial_\theta E_\theta \quad \longrightarrow \quad \theta_{n+1} = \theta_n - \eta \mathbf{S}^{-1} \partial_\theta E_\theta$$

Sidenote

$$\partial_t p_\theta = \mathcal{F} p_\theta$$

$$\mathcal{F} p_G = 0$$



?

Only useful if the Ansatz is autoregressive though

Variational principle

$$\dot{\theta} = \arg \min_{\dot{\eta}} D_{\text{KL}}(p_{\theta + \delta t \dot{\eta}} \| e^{\delta t \mathcal{F}} p_\theta)$$



Stochastic-reconfiguration (SR) equation

$$\dot{\theta} = \mathbf{S}^{-1} \mathbf{F}$$

Fisher metric

$$S_{kk'} = \overline{(O_k^* - \overline{O_k^*})(O_{k'}^* - \overline{O_{k'}^*})}$$

$$O_k(x) := \partial_{\theta_k} \log p_\theta(x)$$

$$\mathbf{F}_k = \overline{(O_k - \overline{O_k})(\mathcal{F}_{\text{loc}} - \overline{\mathcal{F}_{\text{loc}}})}$$

Forces $\sim$ free energy gradient

SR rescales the gradient according to the physical effect of each parameter on the variational probability distribution.

$$\theta_{n+1} = \theta_n - \eta \partial_\theta F_\theta \quad \longrightarrow \quad \theta_{n+1} = \theta_n - \eta \mathbf{S}^{-1} \partial_\theta F_\theta$$

Neural quantum states

G. Carleo and M. Troyer, Science 335(6325), 602-606 (2017)

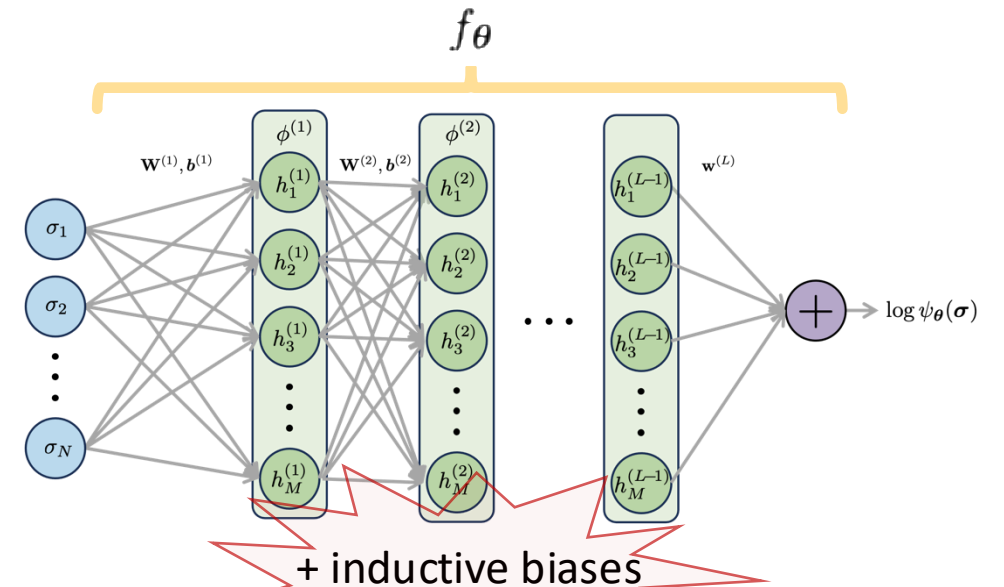
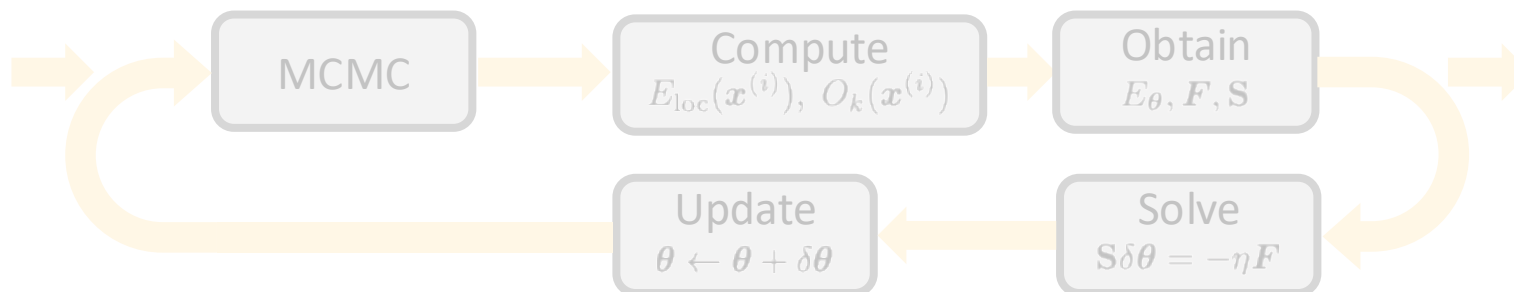
Neural quantum state

$\log \psi_{\theta}(\mathbf{x}) = f_{\theta}(\mathbf{x})$ $\leftarrow$ Work in log space

e.g. L-layer perceptron $f_{\theta}(\mathbf{x}) = \mathbf{w}^{(L)} \cdot \mathbf{h}^{(L-1)}$.

$$\mathbf{h}^{(\ell+1)} = \phi^{(\ell)}\left(W^{(\ell)}\mathbf{h}^{(\ell)} + \mathbf{b}^{(\ell)}\right), \quad \mathbf{h}^{(0)} = \mathbf{x},$$

All the VMC machinery remains the same:



`pip install --upgrade netket`

F. Vicentini *et al.*, SciPost Phys. Codebases, 7 (2022)

What makes a good variational wavefunction?

Neural quantum state

$$\log \psi_{\theta}(\mathbf{x}) = f_{\theta}(\mathbf{x}) \quad \leftarrow \text{Work in log space}$$

Some important qualities

- Efficient
- Optimisable
- Expressive
- Physics-aware

Simplest inductive bias

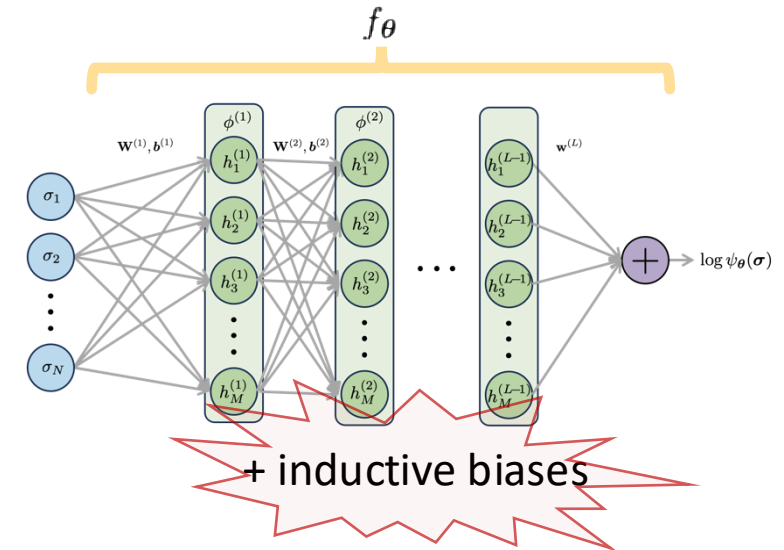
Suppose we already have some cheap *prior*,
e.g. mean field

$$|\phi_{\eta}\rangle \sim |\psi_0\rangle \quad \text{Exact ground state}$$

Then, just add the log values

$$\log \psi_{\theta}(\mathbf{x}) = f_{\theta}(\mathbf{x}) + \log \phi_{\eta}(\mathbf{x})$$

$$|\psi_{\theta}\rangle = e^{f_{\theta}(\hat{\mathbf{x}})} |\phi_{\eta}\rangle$$



Symmetries

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle$$

$$[\hat{U}, \hat{H}] = 0$$

$$\hat{U} |\psi_n\rangle = \chi_n |\psi_n\rangle$$

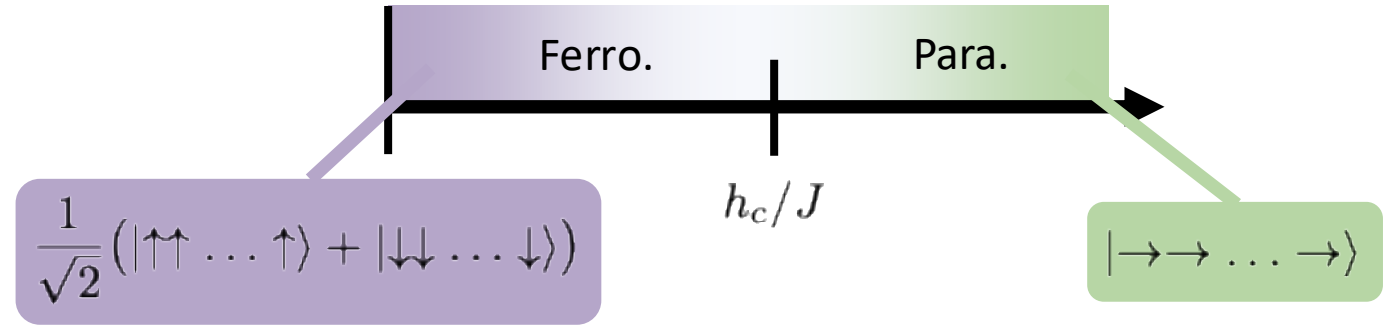
We can constrain the NQS to belong
to the same symmetry sector as the
ground state

Symmetry projection

Why bother with symmetry?

Consider the simplest example

$$\hat{H}_{\text{TFI}} = -J \sum_i \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z - h \sum_i \hat{\sigma}_i^x$$



Two symmetries

Global $\mathbb{Z}_2$ symmetry

$$\sigma \leftrightarrow -\sigma$$

Translational symmetry (PBCs)

$$|\sigma\rangle \leftrightarrow \hat{T}^n |\sigma\rangle, \forall n$$

$$\hat{T} |\downarrow\uparrow\uparrow \dots \uparrow\rangle = |\uparrow\downarrow\uparrow \dots \uparrow\rangle$$

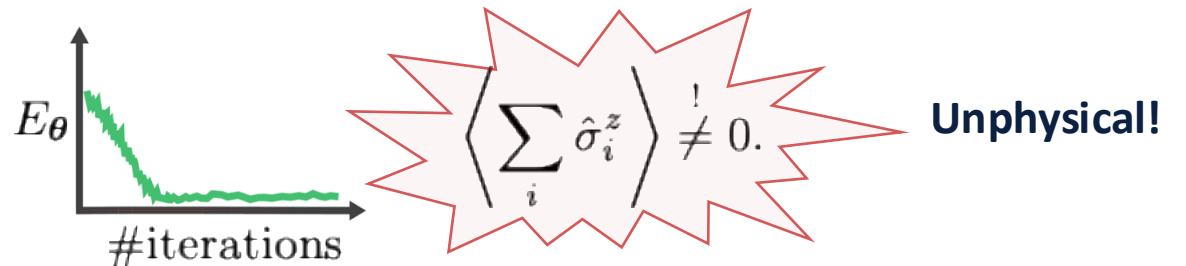
2 Very good inductive bias

Non symmetric:

ψ_θ learns: configurations $\mapsto$ complex number

1 Now let us run VMC in the ferromagnetic regime:

$$\begin{aligned} |\psi_{\theta(0)}\rangle &\xrightarrow{\text{yellow arrow}} |\psi_{\theta(+\infty)}\rangle \propto \psi_{\theta(0)}(\uparrow \dots \uparrow) |\uparrow \dots \uparrow\rangle \\ \theta(0) \text{ random} &\quad + \psi_{\theta(0)}(\downarrow \dots \downarrow) |\downarrow \dots \downarrow\rangle \end{aligned}$$



Symmetric

orbits $\mapsto$ complex number

Invariance and equivariance

General prescription

$$\psi_{\theta}^{(\chi)}(x) = \sum_{g \in G} \chi(g)^* \psi_{\theta}(gx)$$

Network gets evaluated
|G| many times...

Parity

$$\psi_{\theta}^{(+)}(\sigma) = \psi_{\theta}(\sigma) + \psi_{\theta}(-\sigma)$$

Translations

$$\psi_{\theta}^{(k)}(\sigma) = \sum_{n=0}^{N-1} e^{-ikn} \psi_{\theta}(T^n \sigma)$$

The network is evaluated N times...

Can we do better? How to build an invariant (symmetric) network?

Start from two types of building blocks

$$f_{\theta}^{\text{inv}}(gx) = f_{\theta}^{\text{inv}}(x)$$

$$h_{\theta}^{\text{equiv}}(gx) = "g" h_{\theta}^{\text{equiv}}(x)$$

Composable

General construction

$$\log \psi_{\theta}^{\text{inv}}(\sigma) = f_{\theta}^{\text{inv}} \circ h_{\theta}^{\text{equiv}(L)} \circ \dots \circ h_{\theta}^{\text{equiv}(1)}(\sigma)$$

One can do momentum spectrography

Example: convolutional neural quantum states

$$h_{i\alpha}^{(\ell+1)} = \phi \left(\sum_{\beta=1}^{C_{\ell}} \sum_{r=-R}^R W_{\alpha\beta r}^{(\ell)} h_{i+r,\beta}^{(\ell)} + b_{\alpha}^{(\ell)} \right)$$

$$h^{(\ell)}(T\sigma) = Th^{(\ell)}(\sigma)$$

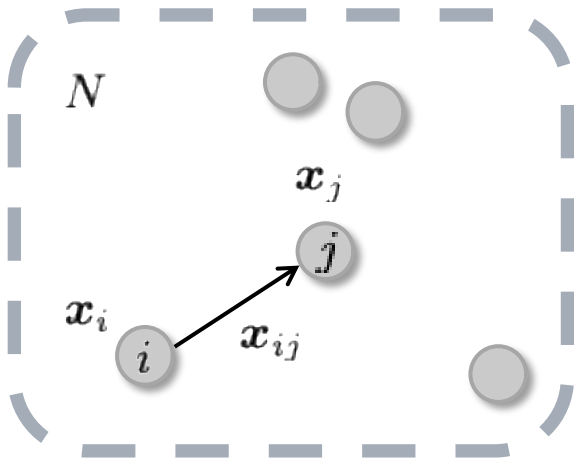
$$\log \psi_{\theta}^{k=0}(\sigma) = \sum_{\alpha=1}^{C_L} w_{\alpha} \left(\frac{1}{N} \sum_{i=1}^N h_{i\alpha}^{(L)} \right)$$

complex

invariant

single forward pass

Simulating interacting particles



N interacting indistinguishable bosons

$$\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N \times d}$$

$$H = - \sum_i \frac{1}{2} \nabla_{\mathbf{x}_i}^2 + \sum_{i,j} V(x_{ij})$$

VMC in the continuum

$$E_\theta = \frac{\int d\mathbf{X} \psi_\theta^*(\mathbf{X}) H(\mathbf{X}) \psi_\theta(\mathbf{X})}{\int d\mathbf{X} |\psi_\theta(\mathbf{X})|^2} = \int d\mathbf{X} p_\theta(\mathbf{X}) E_{\text{loc}}(\mathbf{X})$$

$$E_{\text{loc}}(\mathbf{X}) = -\frac{1}{2} \sum_{i=1}^N \sum_{\mu=1}^d \left((\partial_{x_{i\mu}} \log \psi_\theta(\mathbf{X}))^2 + \partial_{i\mu}^2 \log \psi_\theta(\mathbf{X}) \right) + \sum_{i < j} V(x_{ij})$$

$$\text{MCMC: } \mathbf{X}' | \mathbf{X} \sim \mathcal{N}(\mathbf{X}, \epsilon^2 \mathbf{1})$$

Bosonic quantum statistics

$$\psi(\mathbf{x}_i, \mathbf{x}_j) \stackrel{!}{=} \psi(\mathbf{x}_j, \mathbf{x}_i)$$

Not an inductive bias; actual constraint on the Hilbert space



Naïve idea: symmetry projection

$$\psi_\theta^{\text{Sym}}(\mathbf{X}) = \sum_{\pi \in S_N} \psi_\theta(\pi \mathbf{X})$$

$$|S_N| = N!$$

Intractable

Better idea: use equivariance (deep sets, complete GNNs)

$$\log \psi_\theta(\mathbf{X}) = \rho \left(\sum_{i=1}^N \phi_i(\mathbf{X}) \right)$$

Invariant pooling

Permutation equivariant

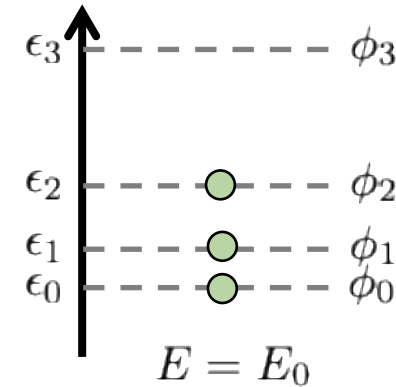
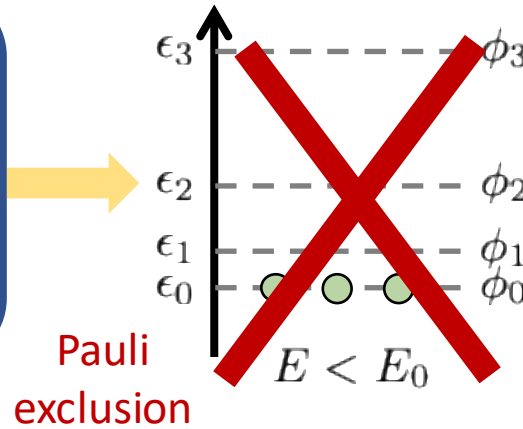
Simulating fermions

Fermionic quantum statistics

$$\psi(\mathbf{x}_i, \mathbf{x}_j) \stackrel{!}{=} -\psi(\mathbf{x}_j, \mathbf{x}_i)$$

Not an inductive bias; actual constraint on the Hilbert space

Pauli exclusion: consider free fermions



$$\log \psi(\mathbf{X}) =$$

$$\begin{pmatrix} \phi_1(\mathbf{x}_1) & \cdots & \phi_1(\mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ \phi_N(\mathbf{x}_1) & \cdots & \phi_N(\mathbf{x}_N) \end{pmatrix},$$

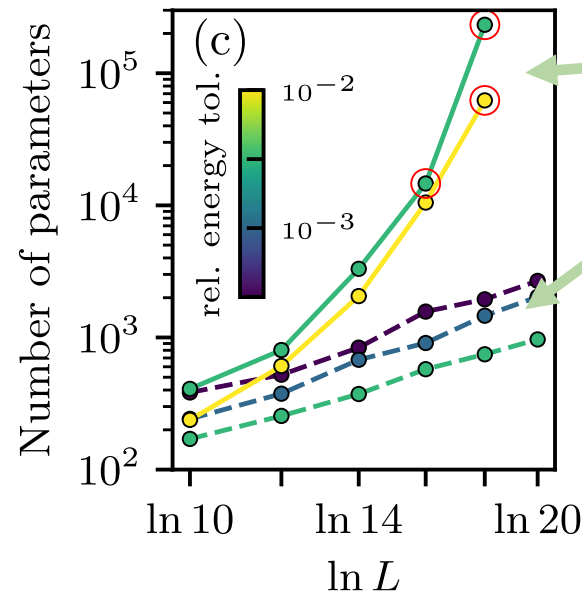
Antisymmetric

Complex sign structure:

$$\psi_0(\mathbf{X}) = \sigma(\mathbf{X}) |\psi_0(\mathbf{X})|$$

Configuration-dependent sign

Very hard to learn from scratch



$\log \psi_{\theta}$ “=” FFN_{θ}

$\log \psi_{\theta}$ “=” $\det(\text{FFN}_{\theta})$

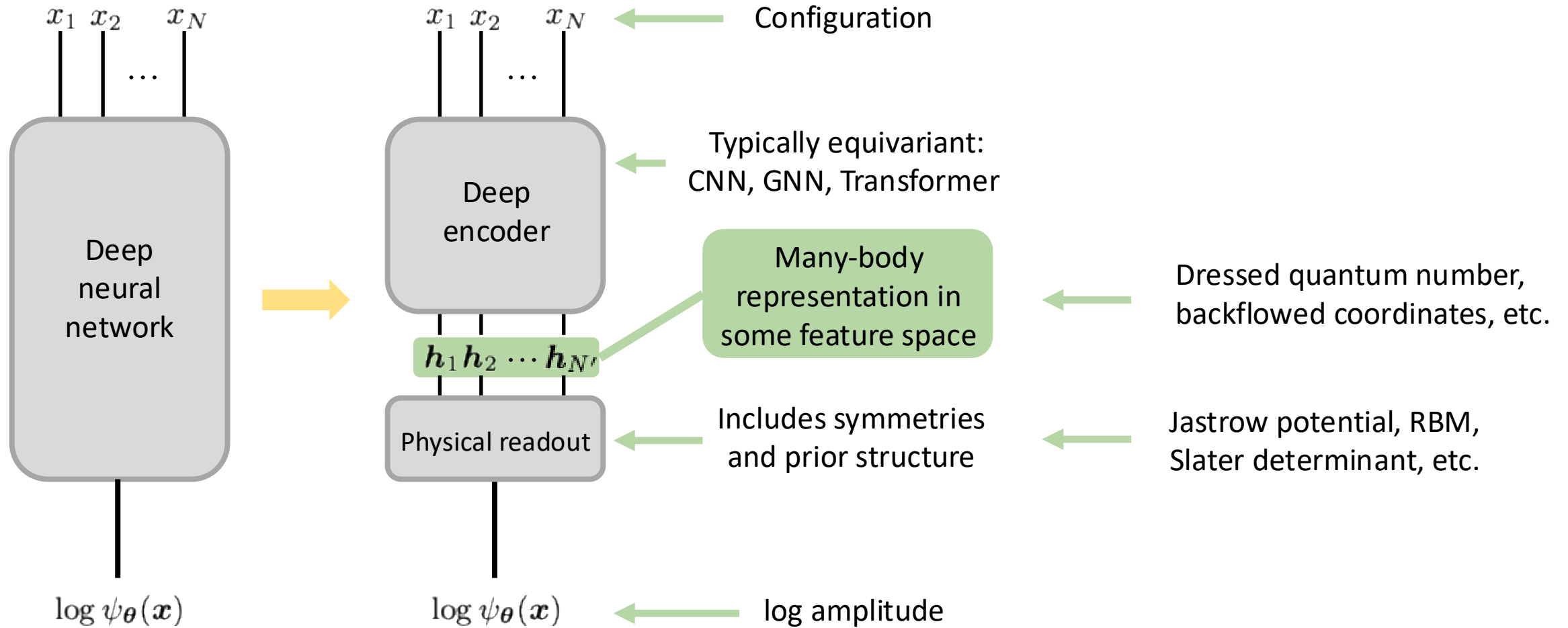
Neural backflow prescription

$$\psi_{\theta}(\mathbf{X}) = \det[\phi_i(\tilde{\mathbf{x}}_j(\mathbf{X}))]$$

$$\tilde{\mathbf{x}}_j(\mathbf{X}) = \mathbf{x}_j + \mathbf{h}_{\theta,j}(\mathbf{X})$$

Many-body effects enter through a dressing of the bare coordinates

General trend



Conclusion

Variational Monte Carlo

- Sampling
- Optimisation
- SR/natural gradients

ML architectures

- Flexible
- Expressive
- Trainable

Problem structure

- Symmetries
- Priors
- Physics

NQS

Check out the lecture notes
github.com/Z-Denis/cecam-nqs

Any questions?